SOFC synthetic data · readiness check · 14 October 2026

Are the stage map and the stage rules enough to make the fake data?

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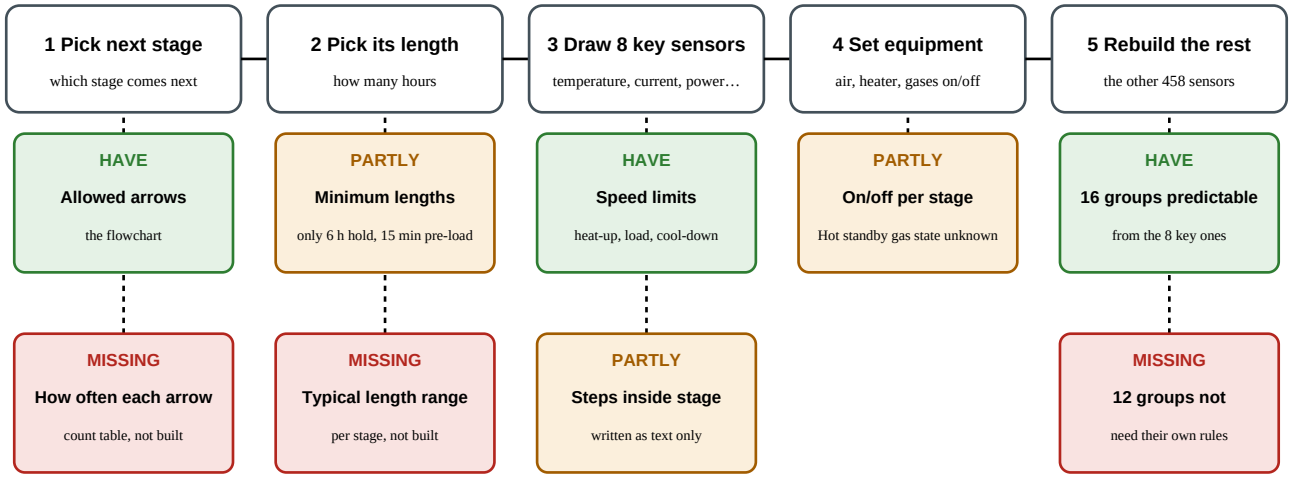
Short answer: enough to decide the order of the stages, not enough to fill them. The stage map says which stage may follow which, and the rules say how to recognise each stage. A fake-data machine also needs to know how long each stage lasts, how often each path is taken, what happens step by step inside a stage, and how the other 458 sensors behave. Most of that can be measured from the 39 days. Five points need your decision.

What each step of the machine needs, and whether we have it



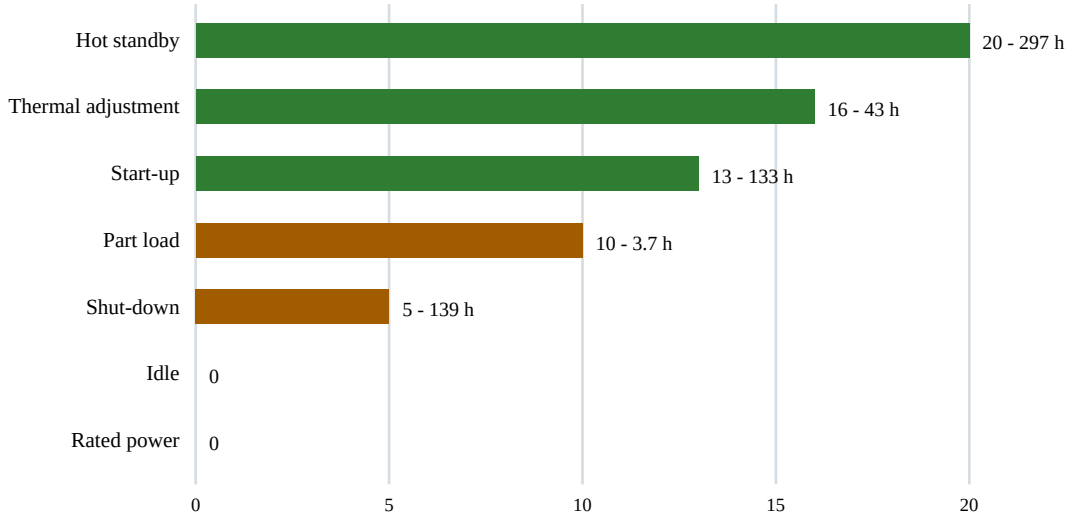
Green comes from the approved stage map and rules; amber is there in part; red is missing.

Already enough for	Not yet enough for
Labelling every minute of the real record (Step 1 done). Refusing impossible orders, for example load straight after a shut-down. Speed limits: heat-up about 50 °C/h (limit 120), load change up to 0.9 A/min, cool-down 45–75 °C/h.	How long each stage lasts and how often each path is taken. The steps inside a stage as rules. Rated power and higher part load (never happened). 12 sensor groups the 8 key sensors cannot predict.

If you remember one thing. The stage map is the skeleton of the fake data. It is correct and finished, but the machine still needs lengths, odds, inside-stage steps and the other sensors before it can make a believable day.

2 Stage by stage

The rules tell the machine what a stage is; the real examples tell it what a stage looks like. Three stages have plenty, two are thin, and two have none.



Usable complete examples (and total hours) in 39 days. Source: Synthetic data generation/library/episodes.csv.

Stage	Usable	Typical length	Verdict	What this means
Hot standby	20	0.7–56 h (median 11 h)	Enough	Enough examples. Two kinds: a hold at about 510 °C and one at 575–610 °C before load.
Thermal adjustment	16	0.5–5.6 h (median 2.7 h)	Enough	Enough. Re-heats and cool-backs of a hot stack with no load.
Start-up	13	1.3–27 h (median 9.6 h)	Enough	Enough, but split into cold and warm starts (8 and 7 in the stage study).
Part load	10	8–44 min (median 22 min)	Thin	Short runs, at most 16 A and 34 kW. 8 of 10 ended in a safety trip, so a calm ramp-down is barely seen.
Shut-down	5	13–59 h (median 21 h)	Thin	Only 5 complete ones; the rest are cut by day-file edges. Forced and natural cooling both appear.
Idle	0	–	None	Only on 2025-10-30 (first file). Nothing usable to learn from.
Rated power	0	–	None	Never reached (maximum 67 % of rated). Can only be made from design values.

Paths between stages: 18 allowed, 10 seen (74 times). Copying the real odds would never produce an allowed but unseen path, such as Part load back to Hot standby or anything with Rated power, so the odds need a rule: real counts plus a small chance for unseen paths, or no chance at all.

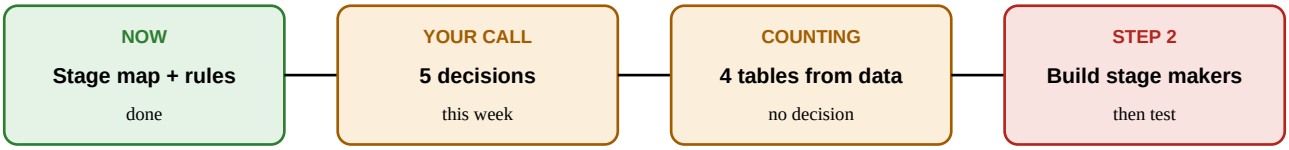
From	Goes next to (times seen)
Hot standby	Thermal adjustment 11 · Part load 10 · Shut-down 4
Thermal adjustment	Hot standby 16 · Shut-down 2

From	Goes next to (times seen)
Part load	Thermal adjustment 7 · Shut-down 3 · Hot standby 0
Start-up	Hot standby 10 · Shut-down 2 and Idle 1 (aborted, outside the map)
Shut-down	Start-up 10

If you remember one thing. Hot standby, Thermal adjustment and Start-up can be learned from the data. Part load and Shut-down are thin, and Idle and Rated power cannot be learned at all, so they need your decision.

3 What to settle next

None of the gaps needs a change to the stage map. The decisions cover what the 39 days never showed; the counting jobs only measure what they already show.



Decisions that are yours

Decision	The question	Suggested default
Rated power (D3)	Never reached. Make it from design values (50 kW AC, 30 A per string) and mark it "not learned", or leave it out.	Leave it out of version 1
How high Part load may go	Seen only up to 16 A and 34 kW. Going higher means guessing from a straight-line stretch (about 51 kW at 30 A).	Stay within 16 A
How a Part load run ends	8 of 10 real runs ended in a safety trip. Should fake runs end with a calm ramp-down, or sometimes with a trip?	Calm ramp-down; trips only behind the fault switch (D12)
Aborted start-ups	Start-up → Shut-down was seen twice and Start-up → Idle once. Allow these paths in fake data?	Allow Start-up → Shut-down, rarely
Idle	Only one real example (the first file). Keep it in fake data?	Leave out; every fake file starts in Shut-down at ambient

Counting jobs that need no decision

Table	What it holds
Odds table	How often each path is taken, with the rule for unseen paths.
Length table	The range of hours per stage and per kind (cold or warm start, 510 °C or 590 °C hold).
Inside-stage scripts	The steps of a Start-up and a Shut-down written as rules with their trigger temperatures.
Equipment table	Air, heater, nitrogen, hydrogen and natural gas on or off in each stage, read from the 12 hard sensor groups.

Points that do not block the fake data

30 K or 40 K near-ambient margin: it only moves a few hours between two parts of Shut-down. The leaking stack in late March: fake data copies it, so note it in the release notes. Purpose of the 510 °C hold: the machine can copy it without knowing why it happens.

If you remember one thing. Settle the five decisions (defaults are ready), let the four tables be counted from the data, and then the stage map plus these additions is enough to start making fake data.